Autonomous Systems Risks

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 25: AI Risk Management

1. Why Autonomous Systems Risk Is Different in Kind

The risks covered in most AI governance frameworks — bias, data leakage, reputational damage, performance degradation — are information-domain risks. Autonomous systems risk is different in kind: when an AI system controls a physical actuator, errors leave the information domain and become physical events. A vehicle veers into a pedestrian. A surgical robot makes an unintended incision. A grid control system destabilizes power distribution. An autonomous drone misidentifies a target. Information-domain errors can typically be corrected, reversed, or mitigated; physical-domain errors may be irreversible. This irreversibility changes the risk calculus fundamentally: the cost of a false negative in a fraud detector is financial loss; the cost of a false negative in an autonomous vehicle's pedestrian detection system may be a life. Governance of autonomous AI must be proportional to these stakes, which is why autonomous systems are the primary targets of the most stringent requirements in EU AI Act, NIST AI RMF, and sector-specific safety standards like ISO 26262 and IEC 61508.

2. The Autonomy Spectrum: SAE J3016 Levels

SAE Level	Name	Human Role	AI Role	Key Risk
0	No automation	Full control	No driving function	None from automation
1	Driver assistance	Monitors and controls all; one system assists	Assists single function (steering OR braking)	Driver over-reliance on single assist
2	Partial automation	Monitors at all times; supervises	Controls steering AND acceleration/braking in specific conditions	Complacency; driver stops monitoring
3	Conditional automation	Must be ready to take over on request	Handles all driving in specific ODD; requests handoff when needed	Handoff problem — human not engaged when takeover needed
4	High automation	Not required within defined ODD	Fully autonomous within ODD; handles failures within ODD	ODD boundary failures; edge cases at ODD limits
5	Full automation	Not required in any condition	Fully autonomous in all conditions	Full system reliability requirement; no human fallback

Level 3 (conditional automation) is the most governance-intensive level because it creates a hybrid human-AI control arrangement where neither the human nor the AI is fully responsible at any given moment. The handoff problem — the failure mode unique to Level 3 — is discussed in depth below.

3. The Handoff Problem: Level 3's Fundamental Challenge

At Level 3 autonomy, the human operator is not actively performing the driving or control task. Their attention drifts to other activities. When the autonomous system encounters a situation it cannot handle — an edge case in its operational design domain (ODD), a sensor failure, an unusual environmental condition — it requests a handoff: 'please take control now.' The human must transition from non-engaged passenger to active operator in seconds. Research consistently shows that this transition takes 3 to 10 seconds, during which a vehicle traveling at 100 km/h covers 83 to 278 meters with degraded or absent human control. The handoff problem is not solved by better AI — it is a fundamental psychomotor and situational awareness challenge. Addressing it requires design choices, not just software improvements:

- Alerting mechanisms that initiate handoff significantly before the autonomous system reaches its competence boundary, providing maximum transition time.
- Automated safe fallback behaviors during the transition period — the vehicle should be slowing and preparing to stop while the human regains awareness.
- **Situational awareness interfaces:** interface design that gives the operator a rapid mental model of the current driving environment before they are expected to control it — not a blank screen until the warning sounds.
- Minimum transition time requirements specified in the system's ODD definition — the system should not request handoff in situations where the available transition time is known to be insufficient.

4. Safety Engineering Standards for Autonomous AI

Safety-critical autonomous systems draw on decades of safety engineering practice developed in aerospace, nuclear, and industrial process control. The primary applicable standards include:

Standard	Domain	Key Requirement
ISO 26262	Automotive functional safety	ASIL (Automotive Safety Integrity Level) assignment; HARA (Hazard Analysis and Risk Assessment); systematic validation
IEC 61508	General electrical/electronic programmable systems	SIL (Safety Integrity Level) assignment; fail-safe design; redundancy requirements
DO-178C	Airborne software	Software level (DAL A-E) based on failure consequence; requirements traceability; coverage-based testing
IEC 62443	Industrial control systems / OT security	Security levels for process control environments where autonomous AI is increasingly deployed

The critical challenge for AI in safety-critical contexts is that these standards were designed for deterministic software. ML-based systems are probabilistic — the same input can produce different outputs across runs, and the system's behavior on out-of-distribution inputs cannot be fully specified at design time. Bridging this gap is one of the unsolved challenges in autonomous AI deployment: EASA (the European Aviation Safety Agency) issued guidelines in 2021 explicitly acknowledging that existing certification frameworks do not fully accommodate ML-based systems.

5. Real-World Incident: Uber ATG Fatality, Tempe Arizona (2018)

Real-World Incident — Uber Advanced Technologies Group, Tempe AZ (2018): A pedestrian pushing a bicycle was struck and killed by an Uber autonomous test vehicle operating in autonomous mode. The vehicle's perception system detected the pedestrian but classified her as an unknown object, then a vehicle, then a bicycle before impact. The system's design suppressed emergency braking for 1.3 seconds to reduce 'false positives.' The safety driver was distracted and did not intervene. The NTSB investigation identified multiple contributing factors: inadequate safety risk assessment, a safety system design that prioritized smooth operation over conservative safe defaults, inadequate operator oversight of the safety driver, and over-reliance on the autonomous system. The case demonstrates that autonomous systems failures result from system-level governance failures, not just technology failures — risk assessment, human oversight design, and safe default behaviors all contributed to the outcome.

6. Physical Adversarial Attacks Against Autonomous Systems

Adversarial attacks against autonomous systems operate in the physical domain as well as the digital one. A 2017 study by Eykholt et al. demonstrated that adding carefully designed stickers to a stop sign — invisible to humans as anything but vandalism — caused a deep learning vision system to classify the sign as a 45 mph speed limit sign with high confidence. Subsequent research extended this to:

- Lane marking adversarial attacks — projecting images onto roadways that cause lane-keeping systems to steer into adjacent lanes.
- Traffic signal adversarial attacks — modifying signal displays in ways human drivers do not notice but AV perception systems misclassify.
- Pedestrian detection evasion — patterns on clothing that cause pedestrian detection systems to fail to classify the wearer as a pedestrian.
- LiDAR spoofing — introducing fake point cloud data using laser sources to create phantom objects or remove real ones from the sensor's representation of the environment.

MITRE ATLAS v2.1 classifies physical adversarial examples under AML.T0043 (Craft Adversarial Data) in the physical domain. Defense requires sensor fusion — requiring consistent readings from multiple independent sensor types (camera, LiDAR, radar) before acting on any single sensor's detection — and adversarial robustness training on augmented datasets that include physical adversarial examples.

Sensor Fusion as Defense: A physical adversarial attack that defeats a camera-based system may not defeat a combined camera + LiDAR + radar fusion system. Requiring cross-sensor consistency before acting on a detection significantly raises the cost of physical adversarial attacks against autonomous vehicles.

7. Ethical Decision-Making in Autonomous Systems

When an autonomous vehicle's collision is unavoidable, the system's response encodes an ethical decision — whether or not that decision was made explicitly by the designers. The MIT Moral Machine experiment collected 40 million decisions from participants in 233 countries and found significant cross-cultural variation in ethical preferences for autonomous vehicle collision decisions: preferences for saving more lives vs. saving specific categories of people, and whether passenger safety should be prioritized over bystander safety. A single global system cannot accommodate this variation without explicit localization of ethical parameters. The responsible design posture requires:

- Making ethical choices explicit — document the ethical framework (utilitarian, egalitarian, passenger-priority) that underlies collision response design.
- Subjecting those choices to independent ethical review before deployment.
- Communicating the ethical framework transparently to users, regulators, and the public.
- Assessing cross-jurisdictional variation in ethical and legal requirements — what is required or permitted in Germany may differ from Japan or California.

For autonomous weapons systems, the doctrine of proportionality and discrimination (core principles of the law of armed conflict) apply with full force — the system must be able to distinguish combatants from non-combatants and must not cause disproportionate collateral damage.

8. Accountability Gaps: Who Is Responsible When AI Causes Harm?

Traditional liability frameworks assume identifiable human decision-makers in the causal chain. Autonomous systems break this assumption. When an autonomous vehicle causes a fatality, liability may rest with the vehicle manufacturer, software developer, fleet operator, human owner, sensor vendor, or regulatory authority. Different parties have different information and different levels of control; the causal chain from training data choices to a specific failure may be extremely difficult to reconstruct post-incident. Current and emerging frameworks include:

Framework	Jurisdiction	Liability Approach
Product liability law	US	Manufacturers liable for defective products without proof of negligence; extends to AI systems as products
EU AI Act Article 17	EU	Strict liability framework for high-risk AI systems; shifts burden of proof toward AI deployers
EU Product Liability Directive (proposed amendments)	EU	Extends product liability to software including AI; removes software exemption
DoD Directive 3000.09	US Defense	Requires 'appropriate levels of human judgment over the use of force' for autonomous weapons; no bright-line definition

The accountability landscape is shifting toward clearer assignment of liability for autonomous system harms. Organizations deploying autonomous systems should assume liability exposure under product liability frameworks and design documentation, audit trails, and incident response accordingly.

9. Human Oversight Design Requirements

Effective human oversight of autonomous systems is not a policy statement or a monitor screen. EU AI Act Article 14 mandates human oversight for high-risk AI systems with specific requirements that must be implemented as product features:

- Operators must be able to understand the system's capabilities and limitations — not at a general level, but specifically enough to recognize when the system is operating outside its competence.
- Operators must be able to detect failures — situational awareness tools that surface system state, confidence, and anomalies in real time.
- Operators must be able to intervene and override — control interfaces that allow rapid intervention without requiring the operator to first understand what went wrong.
- **Operators must be able to shut down the system:** a shutdown capability that cannot be overridden by the autonomous system itself. An autonomous system that can disable its own shutdown capability is an unacceptable governance risk.

The NIST AI RMF GOVERN function's human oversight requirements additionally address organizational structure: operators must not be incentivized against exercising oversight. If operators are measured and rewarded on throughput and efficiency of autonomous operations, they face pressure not to intervene even when intervention is warranted. Governance must protect the act of intervention, not just the technical capability for it.

10. Automation Complacency and Skill Decay

Automation irony (also called the automation paradox) is the documented phenomenon where increasing automation leads to degraded human performance in the automated task — precisely when that performance is most needed during automation failures. Commercial pilots flying highly automated aircraft have demonstrated reduced manual flying proficiency compared to pre-automation generations. Surgeons using robotic assistance have reduced manual surgical proficiency. Cybersecurity analysts delegating triage to SOAR automation lose threat-hunting proficiency. The problem is structural: if humans are never required to perform the task during normal operations, they lose the proficiency and situational awareness required to perform it during abnormal ones. Organizational responses include mandatory manual operation periods (requiring pilots to hand-fly approaches periodically), proficiency training programs that maintain skills even when automation rarely requires them, and cognitive task analysis to identify what situational awareness operators must maintain even when not actively controlling the system.

11. Testing Autonomous Systems: The Incompleteness Problem

Testing an autonomous system to the level of statistical confidence required for safety-critical certification is a fundamentally unsolved problem. The space of possible scenarios an autonomous system might encounter in the real world is effectively infinite. Waymo's vehicles accumulated more than 20 million real-world miles and 15 billion simulation miles before commercial deployment in San Francisco — and incidents still occurred. Key testing approaches and their limitations:

- Real-world testing — provides highest fidelity but is slow, expensive, and dangerous for rare high-consequence scenarios (you cannot deliberately test emergency braking at highway speeds with a real pedestrian).
- Simulation — covers large portions of the nominal scenario space efficiently, but limited by simulation fidelity; rare real-world conditions not well-modeled in simulation remain blind spots.

- **Staged deployment (ODD expansion):** start with a tightly defined, heavily tested operational design domain; expand only as evidence of safe performance accumulates. This is the industry-standard approach for managing the incompleteness problem.
- Adversarial scenario generation — deliberately constructing test cases designed to find failure modes, rather than relying on random or representative scenario sampling.

12. Regulatory Landscape: Sector-Specific and Evolving

Regulatory frameworks for autonomous systems are sector-specific and geographically fragmented. The compliance question is not 'is this safe?' but 'does this meet the regulatory requirements of every sector and jurisdiction where it operates?' — requirements that are a moving target:

- **Aviation (FAA):** most mature framework; full certification required for autonomous flight functions; process takes years and costs tens of millions. DO-178C and DO-254 are the applicable standards.
- **Ground transportation (US):** no federal autonomous vehicle framework as of 2025; state-by-state rules that vary significantly. NHTSA issues voluntary guidance but has not promulgated mandatory standards.
- **Defense (US):** DoD Directive 3000.09 requires 'appropriate levels of human judgment over the use of force' for autonomous weapons systems, with no bright-line definition of 'appropriate.'
- **EU AI Act (2024):** classifies certain autonomous systems as high-risk, requiring conformity assessments, human oversight provisions per Article 14, and transparency requirements per Article 13.

Key Terms Glossary

Term	Definition
SAE J3016	Standard defining six levels of vehicle automation (0–5) based on the degree to which the AI performs the driving task vs. the human.
Operational Design Domain (ODD)	The specific conditions within which an autonomous system is designed and validated to operate safely — geography, weather, speed range, road type.
Handoff problem	The safety challenge at Level 3 autonomy where a non-engaged human operator must rapidly regain situational awareness and control when the autonomous system requests takeover.
HARA (Hazard Analysis and Risk Assessment)	ISO 26262 systematic process for identifying hazards, estimating risk, and assigning ASIL to automotive system functions.
ASIL (Automotive Safety Integrity Level)	ISO 26262 classification (A–D) of the rigor required to achieve acceptable residual risk for an automotive function; ASIL D is the most stringent.
Physical adversarial attack	Attack that manipulates the physical environment (signs, lane markings, LiDAR sources) rather than digital inputs to cause an autonomous system to misperceive its environment.
Sensor fusion	Combining data from multiple independent sensor types (camera, LiDAR, radar) to produce a more reliable environmental model than any single sensor could provide.

Automation irony	Documented phenomenon where increasing automation degrades human proficiency in the automated task, precisely when that proficiency is most needed during automation failures.
EU AI Act Article 14	Mandates human oversight capability for high-risk AI systems: operators must understand capabilities/limitations, detect failures, intervene, and shut down the system.
DoD Directive 3000.09	US Defense directive requiring 'appropriate levels of human judgment over the use of force' for autonomous and semi-autonomous weapons systems.
Staged deployment	Risk management approach for autonomous systems: deploy within a tightly defined ODD first, expand only as evidence of safe performance accumulates.
AML.T0043 (MITRE ATLAS)	ATLAS technique for crafting adversarial data, including physical adversarial examples that manipulate the environment to cause autonomous system misperception.

Quick Revision Checklist

- Autonomous systems risk is different in kind from information-domain AI risks — physical errors may be irreversible; governance must be proportional to physical harm potential.
- SAE J3016 defines six automation levels (0–5); Level 3 is the most governance-intensive because it creates hybrid human-AI responsibility with a critical handoff problem.
- The handoff problem: humans require 3–10 seconds to regain situational awareness after an automation takeover request; vehicles travel 80–280 meters in that window at highway speeds.
- Safety standards for autonomous AI: ISO 26262 (automotive/ASIL), IEC 61508 (general/SIL), DO-178C (aviation/DAL) — all designed for deterministic software, creating a gap for probabilistic ML systems.
- Physical adversarial attacks manipulate the environment (signs, lane markings, LiDAR) to cause misperception; sensor fusion is the primary defense.
- Ethical decision-making must be explicit, independently reviewed, and communicated transparently — collision response encodes an ethical choice whether or not designers made it consciously.
- Accountability is fragmented across manufacturers, developers, operators, and regulators; EU AI Act Article 17 and proposed Product Liability Directive amendments are moving toward stricter liability.
- EU AI Act Article 14 mandates operator capability to understand limitations, detect failures, intervene, and shut down — these are product-level requirements, not policy statements.
- Automation irony: automation degrades human proficiency in the automated task; maintain skills through mandatory manual operation periods and proficiency training.
- Testing cannot achieve complete coverage of all possible scenarios; staged ODD expansion is the standard approach for managing this incompleteness.
- Regulatory frameworks are sector-specific and geographically fragmented; compliance requires sector-specific analysis in every operating jurisdiction.

10 Exam-Based MCQs — Autonomous Systems Risks

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. ***Scenario:*** *Meridian Robotics is deploying a Level 3 autonomous vehicle system on a commercial fleet of highway freight trucks. During field testing, safety engineers observe that when the system requests a handoff at highway speeds, drivers take an average of 8 seconds to regain effective vehicle control. The system's current design provides a 4-second warning before reaching its operational design domain boundary.* Which design change is MOST CRITICAL to address the identified safety risk?

A) Retrain the perception model on additional highway scenarios to extend the operational design domain boundary.

B) Increase the handoff warning lead time and implement automated speed reduction during the handoff transition period.

C) Require drivers to maintain hands on the wheel at all times to reduce handoff transition time.

D) Upgrade the vehicle's sensor suite to LiDAR + radar fusion to improve edge case detection.

Answer: B **Explanation:** B is correct because the core problem is a mismatch between the 4-second warning and the 8-second human transition time — the handoff window is insufficient for safe human takeover. Increasing warning lead time and implementing automated safe fallback behaviors (speed reduction) during the transition period directly addresses this gap. A extends the ODD but does not address the handoff transition problem when the ODD boundary is eventually reached. C is a partial mitigation (keeping hands on the wheel may reduce transition time somewhat) but does not address the structural gap between warning time and transition time, and may create its own safety concerns. D improves perception but does not address the human-machine control transition problem.

Q2. ***Scenario:*** *The NTSB investigation of an autonomous vehicle fatality found that the vehicle's perception system correctly detected the pedestrian but suppressed emergency braking for 1.3 seconds to reduce false positives. The safety driver was not monitoring the road. The system design had not undergone a formal hazard analysis.* Which safety engineering process, had it been applied, would MOST DIRECTLY have identified the braking suppression design as a hazard requiring mitigation?

A) Model calibration assessment to improve confidence score reliability.

B) HARA (Hazard Analysis and Risk Assessment) following ISO 26262.

C) Freedom-to-operate analysis to assess patent risk in the braking system design.

D) Drift detection monitoring to identify perception model degradation in production.

Answer: B **Explanation:** B is correct because HARA (Hazard Analysis and Risk Assessment) under ISO 26262 systematically identifies hazardous events, estimates their severity and probability, and assigns ASIL requirements to system functions. A formal HARA would have identified 'failure to brake when a pedestrian is detected' as a hazard requiring ASIL-level mitigation, making the 1.3-second braking suppression design a compliance issue requiring redesign. A (model calibration) addresses confidence score reliability, not the safety engineering of braking logic. C (FTO analysis) is an IP risk management tool, not a safety engineering process. D (drift detection) is a production monitoring capability that would not have identified the pre-deployment design flaw.

Q3. ***Scenario:*** *A security researcher publishes a paper demonstrating that applying a specific pattern of retro-reflective tape to a stop sign causes a leading autonomous vehicle manufacturer's camera-based vision system to classify the stop sign as a yield sign with 97 percent confidence. The manufacturer's perception system uses only camera-based detection without sensor fusion.* Which defense would MOST EFFECTIVELY mitigate this category of attack across future deployments?

A) Implementing query rate limiting on the vehicle's API to prevent systematic model probing.

B) Retraining the camera model on adversarial examples including the specific tape pattern.

C) Implementing sensor fusion requiring consistent classification from camera, LiDAR, and radar before acting on any sign detection.

D) Encrypting the vehicle's inference pipeline to prevent attackers from analyzing model outputs.

Answer: C **Explanation:** C is correct because sensor fusion requires agreement across independent sensor modalities — an attack that defeats the camera-based classifier is unlikely to simultaneously fool LiDAR depth-and-shape analysis and radar-based detection in the same way. Sensor fusion dramatically raises the attack cost. B (adversarial training on the specific pattern) is a narrow fix — it addresses this specific attack but not the broader class of physical adversarial examples that could be discovered and applied. A (query rate limiting) is relevant to digital model extraction attacks, not physical adversarial attacks on deployed vehicle sensors. D (encryption of inference pipeline) prevents analysis of model outputs remotely, not physical manipulation of the sensor environment.

Q4. Which SAE J3016 automation level creates the MOST significant governance challenge due to the handoff problem between human and AI control?

A) Level 1 — Driver Assistance

B) Level 2 — Partial Automation

C) Level 3 — Conditional Automation

D) Level 4 — High Automation

Answer: C **Explanation:** C is correct because Level 3 (Conditional Automation) is the only level that expects the human to be fully disengaged from the driving task during normal operation but simultaneously expects the human to be capable of taking over on request within seconds. This creates the handoff problem: humans not engaged in a task lose situational awareness rapidly and require 3–10 seconds to recover it. Level 1 and Level 2 require the human to actively monitor and supervise, so the handoff problem is minimal. Level 4 does not require handoff within the ODD — the system handles all failure modes internally, and humans are not expected to take over within the ODD.

Q5. An autonomous trading system executes thousands of orders per minute with no human review. An attacker who has compromised the system's signal input wants to trigger it to place large destabilizing orders without causing the system's anomaly detection to alert. Which MITRE ATLAS v2.1 technique category BEST describes this attack objective?

A) Model evasion — crafting inputs that cause specific misclassifications while staying below anomaly detection thresholds.

B) Model poisoning — corrupting the training data to embed a persistent backdoor in the model.

C) Model extraction — querying the system to reconstruct its decision logic for offline exploitation.

D) Inference API access — probing the system's API to characterize its response patterns.

Answer: A **Explanation:** A is correct because the attacker's goal is to manipulate the system's outputs (trade orders) in a specific direction while remaining below anomaly detection thresholds — this is model evasion: crafting inputs that cause desired misclassifications while evading detection. The attacker has system access and is generating inputs designed to produce specific controlled outputs. B (model poisoning) corrupts training data, but the scenario describes runtime manipulation of a deployed system, not training data corruption. C (model extraction) aims to replicate the model's behavior externally, not manipulate the live system. D (inference API access) describes probing for characterization, not active exploitation to trigger specific actions.

Q6. Which EU AI Act provision mandates that operators of high-risk AI systems must be able to understand system capabilities and limitations, detect failures, and intervene and shut down the system?

A) Article 9 — Risk management system requirements

B) Article 13 — Transparency and provision of information to users

C) Article 14 — Human oversight

D) Article 17 — Quality management system

Answer: C **Explanation:** C is correct because EU AI Act Article 14 specifically mandates human oversight for high-risk AI systems with explicit requirements: operators must understand capabilities and limitations, detect malfunctions, intervene and override, and shut down the system. These are product-level implementation requirements. A (Article 9) addresses the risk management system the provider must maintain. B (Article 13) addresses transparency and user documentation requirements. D (Article 17) addresses quality management system requirements for providers. Article 14 is the specific human oversight mandate.

Q7. A nuclear plant operator transitions to an AI-assisted control system for reactor coolant management. After 18 months of AI-assisted operation, a drill reveals that operators can no longer manually manage coolant flow as effectively as they could before AI assistance was introduced. Which phenomenon does this BEST illustrate?

A) Model drift — the AI system's performance has degraded as operating conditions changed.

B) Concept drift — the relationship between input parameters and appropriate coolant adjustments has changed over time.

C) Automation irony — automation has degraded the human operators' proficiency in the task the automation performs.

D) Adversarial drift — an attacker has manipulated the AI system to degrade operator confidence.

Answer: C **Explanation:** C is correct because the scenario describes automation irony (also called the automation paradox): operators who are not required to perform the coolant management task during normal AI-assisted operation lose proficiency in that task, precisely when it is most needed during AI system failure. This is a well-documented phenomenon in aviation, surgery, and process control. A (model drift) refers to AI performance degradation, not human skill degradation. B (concept drift) refers to changing relationships in the AI's prediction task. D (adversarial drift) involves deliberate attacker manipulation, not indicated in the scenario.

Q8. Which of the following BEST describes the primary safety engineering challenge of applying ISO 26262 and IEC 61508 to ML-based autonomous AI systems?

A) These standards require physical redundancy, which is too expensive to implement in software-based AI systems.

B) These standards were designed for deterministic software and do not fully accommodate the probabilistic behavior and out-of-distribution uncertainty of ML-based systems.

C) These standards require human certification of every decision made by the system, which is impractical at AI decision rates.

D) These standards apply only to hardware components and do not address software or AI elements.

Answer: B **Explanation:** B is correct because ISO 26262 and IEC 61508 were developed for deterministic software where the same input always produces the same output and system behavior can be fully specified at design time. ML-based systems are probabilistic — behavior on out-of-distribution inputs cannot be fully specified or verified using traditional certification methods like requirements traceability and structural coverage. This gap is explicitly acknowledged by EASA and other regulatory bodies. A is wrong — redundancy requirements in these standards are not limited to physical redundancy and are not prohibitively expensive in principle. C is wrong — these standards do not require human certification of individual decisions. D is wrong — both standards apply to programmable electronic systems including software.

Q9. An AI team is preparing to deploy a Level 4 autonomous warehouse robot system. Risk assessors identify that the system has been extensively tested in a controlled warehouse environment but has not been tested in adjacent areas including loading docks and parking lots. Which deployment approach BEST manages this testing incompleteness risk?

A) Deploy across all areas simultaneously to collect real-world performance data as quickly as possible.

B) Deploy within the tested ODD (warehouse floor only) and expand to adjacent areas only as evidence of safe performance accumulates.

C) Retrain the model on simulation data representing loading dock and parking lot scenarios before any deployment.

D) Deploy in adjacent areas first, since the warehouse environment is riskier due to heavier traffic.

Answer: B **Explanation:** B is correct because staged deployment within a well-defined and tested ODD — expanding only as evidence accumulates — is the industry-standard approach for managing the incompleteness of autonomous system testing. The loading dock and parking lot represent untested operational conditions that should only be entered after additional testing or staged expansion with conservative parameters. A is wrong — deploying across all areas simultaneously exposes the system to untested conditions at full scale before evidence of safe performance in those conditions exists. C (simulation retraining) is a useful complement but does not replace staged real-world deployment with safety monitoring. D reverses the logic — the tested area is safer to deploy in than the untested areas.

Q10. Which US Department of Defense directive specifically addresses human oversight requirements for autonomous weapons systems?

A) DoD Instruction 8500.01 — Cybersecurity

B) DoD Directive 3000.09 — Autonomy in Weapon Systems

C) NIST AI RMF 1.0 — GOVERN function

D) EU AI Act Article 14 — Human Oversight

Answer: B **Explanation:** B is correct because DoD Directive 3000.09 (Autonomy in Weapon Systems) specifically requires 'appropriate levels of human judgment over the use of force' for autonomous and semi-autonomous weapons systems. It is the primary US defense policy document governing human oversight of lethal autonomous systems. A (DoD Instruction 8500.01) addresses cybersecurity requirements for DoD information systems, not autonomous weapons oversight. C (NIST AI RMF) is a civilian framework without authority over weapons systems. D (EU AI Act Article 14) is an EU regulation that does not govern US defense systems.